

Tools for Recovering and Visualizing Intrasite Spatial Data: Interpolation and Spatial Autocorrelation.

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Intra-Site Spatial Data Recovery and Visualization

Historical Issues

- Locations of activities – what happened where and why?
- Locations of different social groups, especially groups among whom there may be conflicts of interest (class, ethnicity, gender). We want to identify, the groups, the conflicts, and the strategies deployed in inter-group competition.
- Sites are not spatially “homogeneous artifact mines”

Sources of Data : Archaeological Spatial Structure

- **Site structure:** patterns in the location of non-portable artifacts... features
 - buildings
 - pits
 - fences
- **Intra-site Artifact Distributions:** patterns in the horizontal distribution of artifacts across sites.... Density surfaces.

Capturing and Analyzing Data on Horizontal Distribution of Artifacts

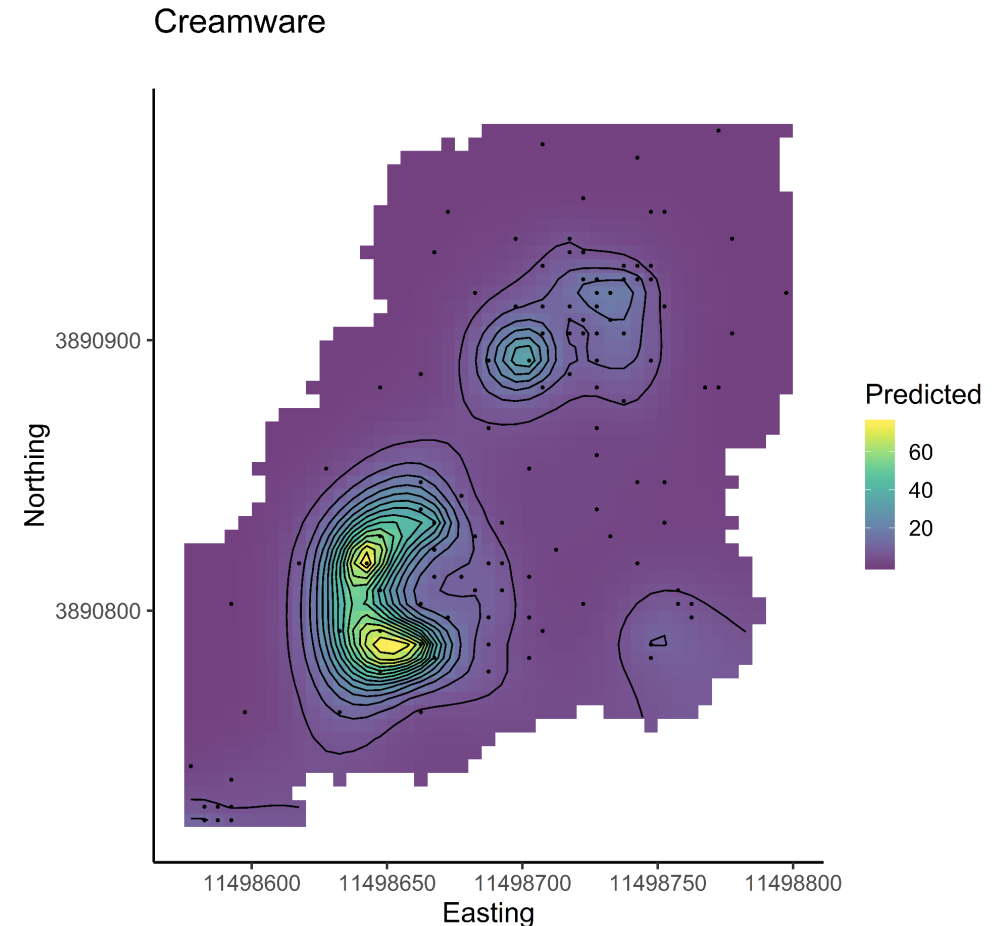
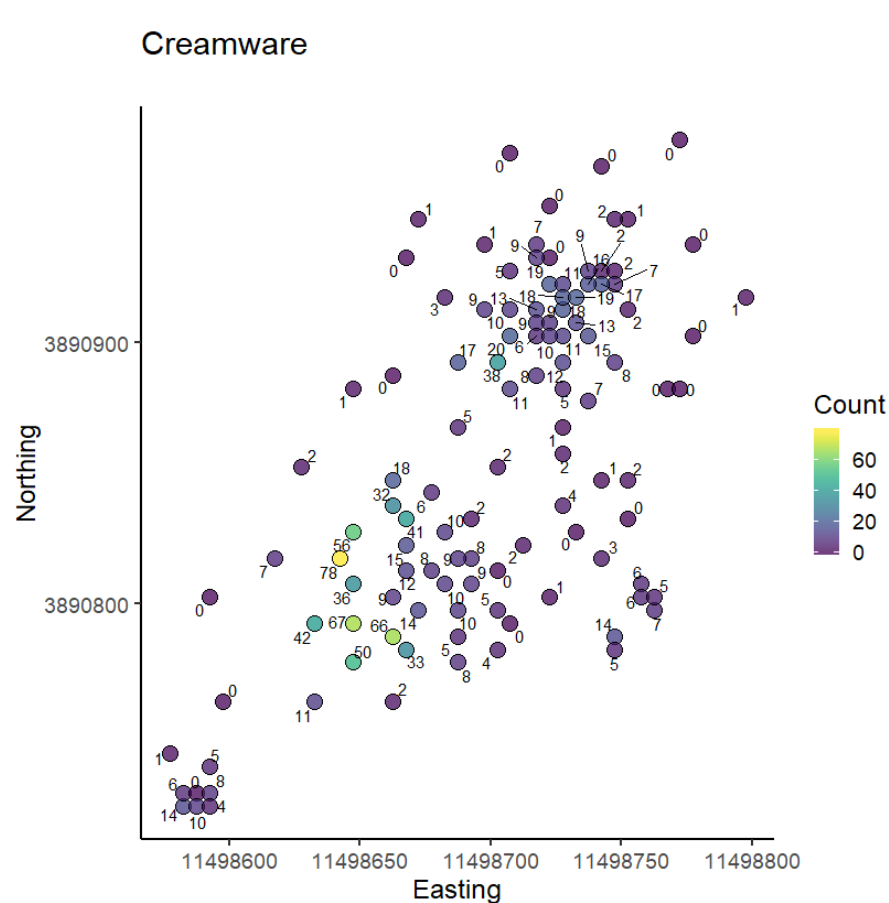
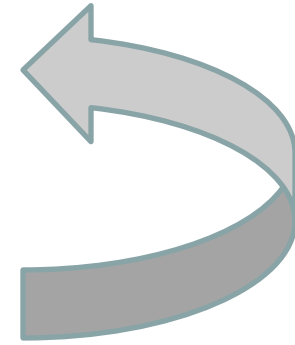
1. We want to model the spatial data recovery process

- Model how we collect spatial data.

2. The analysis process

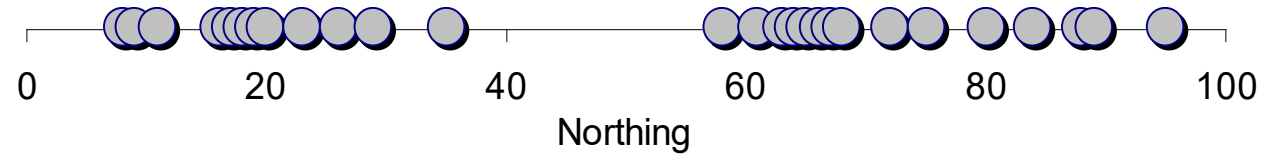
- Map raw data and fit statistical models to the data to help identify patterns and the scale of patterning.

Feedback:
Variogram



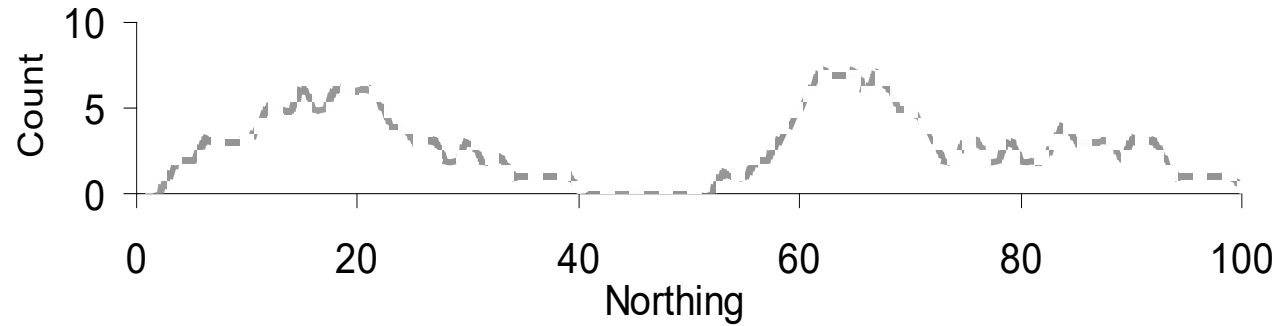
Recovery

The Point Process:



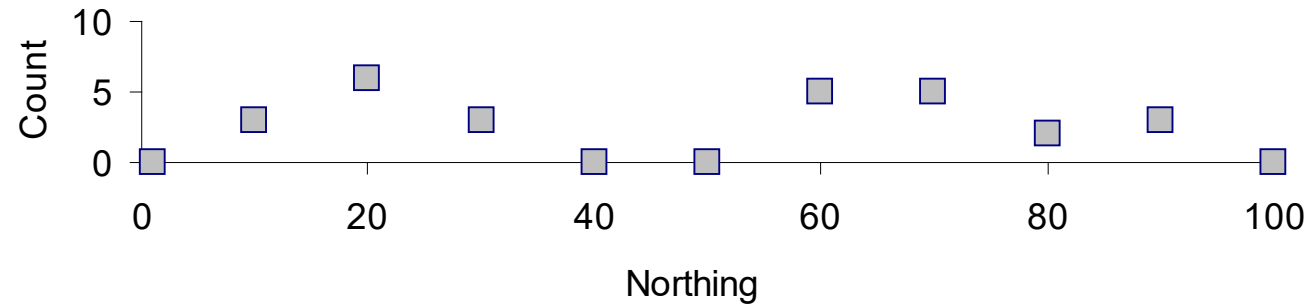
The Moving-Average Process:

(quadrat diameter=10)



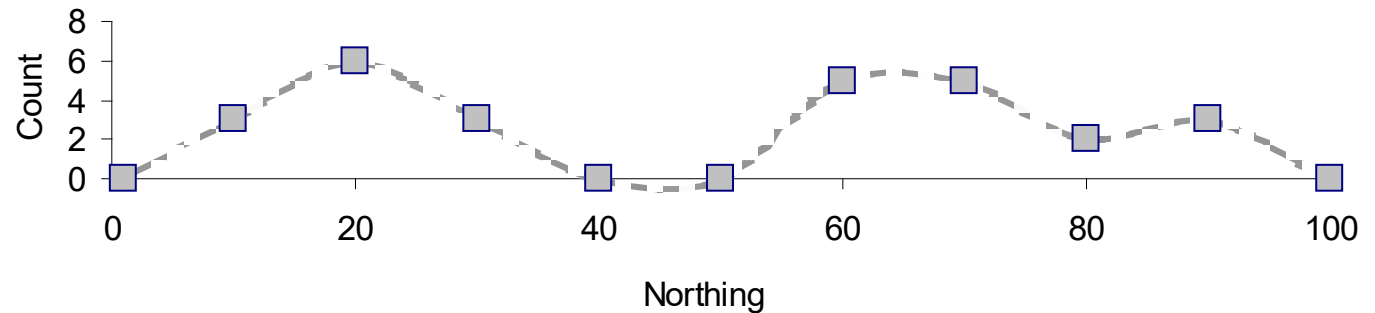
Sample the M-A Process

(quadrat spacing =10)



Analysis

Estimate the M-A Process from the sample:



Important Decisions in Designing a Spatial Sampling Strategy

Quadrat size

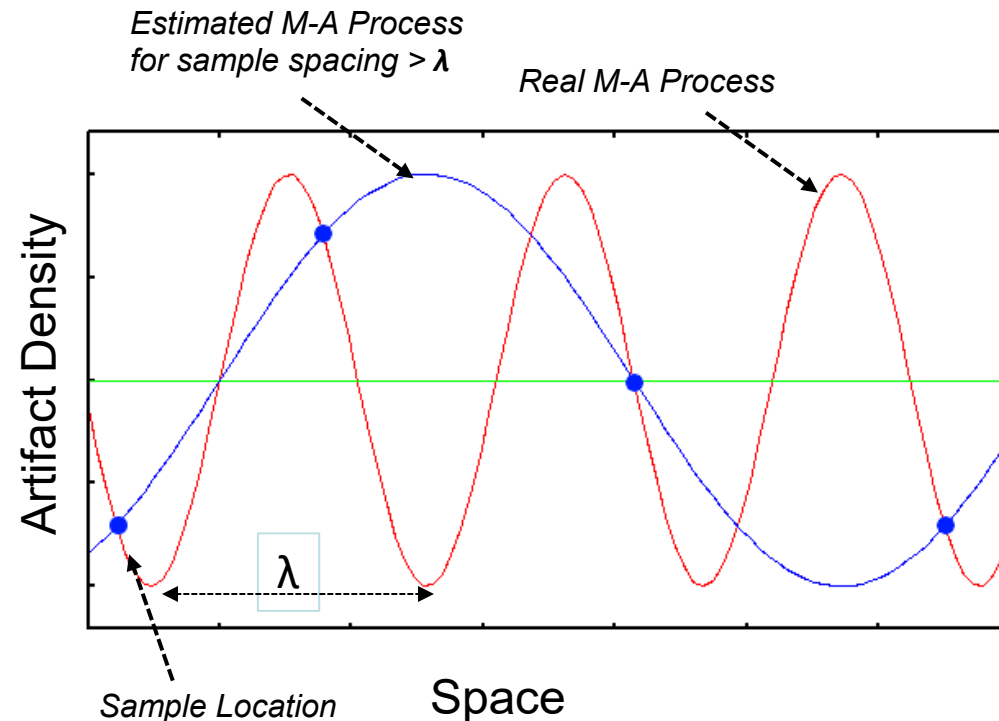
- Determines the minimum possible quadrat spacing
e.g. with 5-foot quadrats, quadrats centers can be no less than 5 feet apart

Quadrat spacing

- Determines the *minimum size* of the of the spatial phenomena we can reliably see.

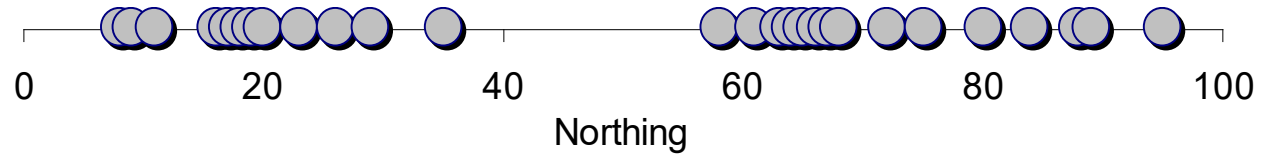
Help from signal processing

- Assume artifact densities (the M-A process) follow a sine wave in 1-D space, with wavelength λ .
- To be able to reconstruct the sine wave from samples, **sample spacing must be $< \lambda/2$** . *e.g.* 10-foot wavelength requires sample spacing less than 5 feet.
- The sine wave is worst-case scenario.



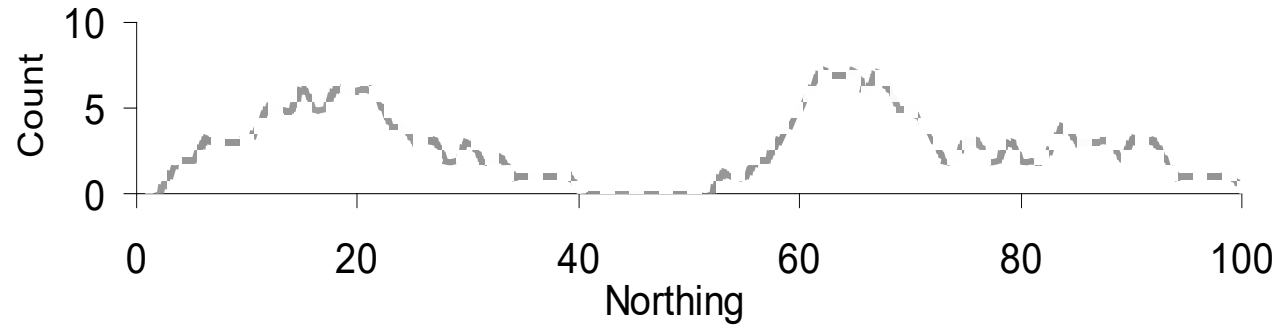
Recovery

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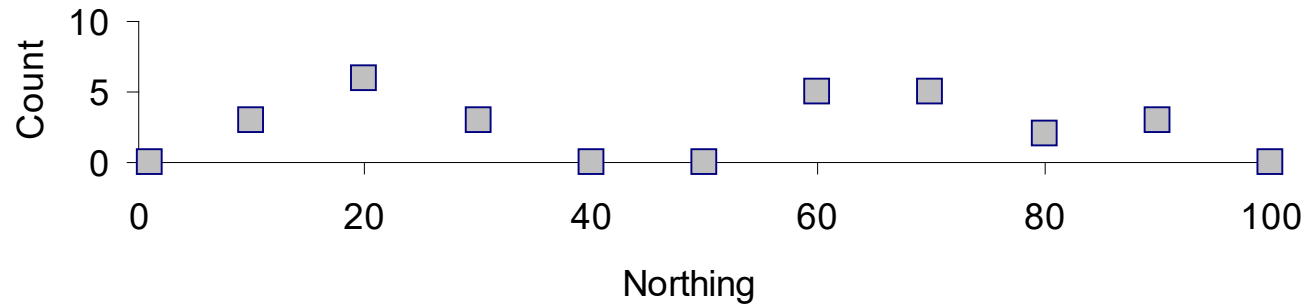
The Moving-Average Process:

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Sample the M-A Process

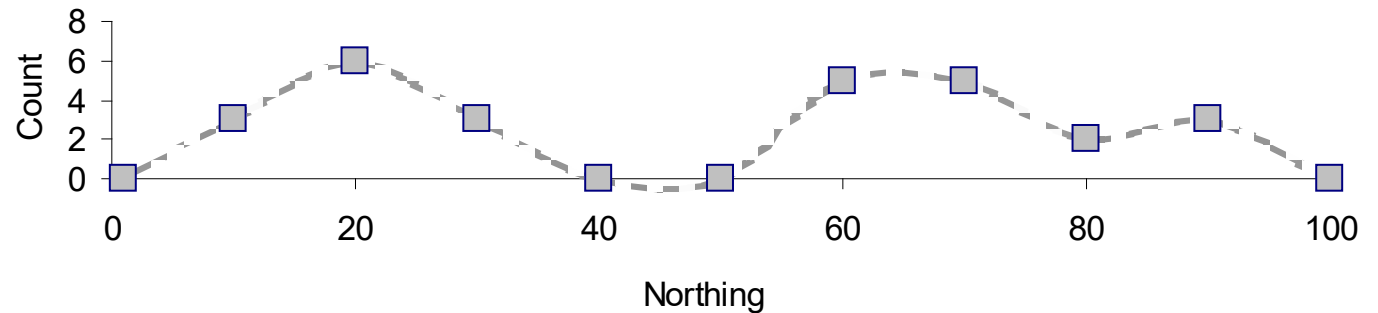
(quadrat spacing =10)



Analysis

Estimate the
M-A Process
from the sample:

Interpolation



Interpolation

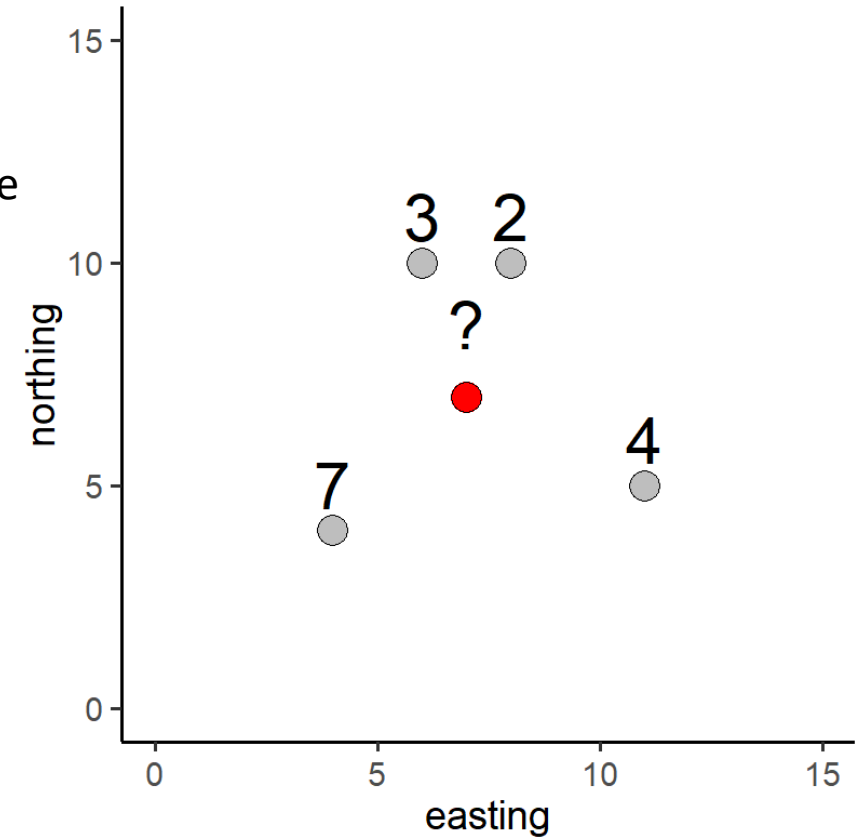
Interpolation is the estimation of unknown z-values at unsampled locations by fitting a continuous surface that passes through (or near) the known z-values at scattered (x, y) sample points — turning isolated observations into a predictions on a grid.

Many methods...

- Inverse distance weighting (IDW)
- Kriging
- Others
 - TINs (triangulated irregular networks),
 - splines
 - polynomial regression (local, global)

IDW and **Kriging** both make estimates of value of the z variable at an unsampled point in (x,y) space, as a ***weighted average of the values at nearby points, where z values are known.***

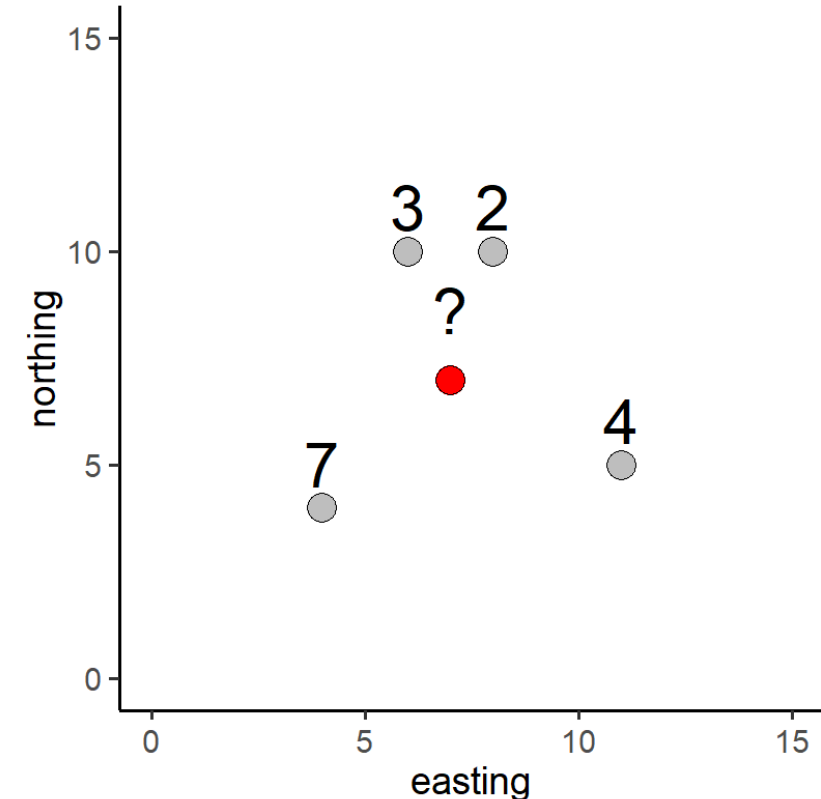
So....



$$\hat{z}_j = \frac{\sum_{i=1}^n w_i z_i}{\sum_{i=1}^n w_i}$$

Inverse Distance Weighting

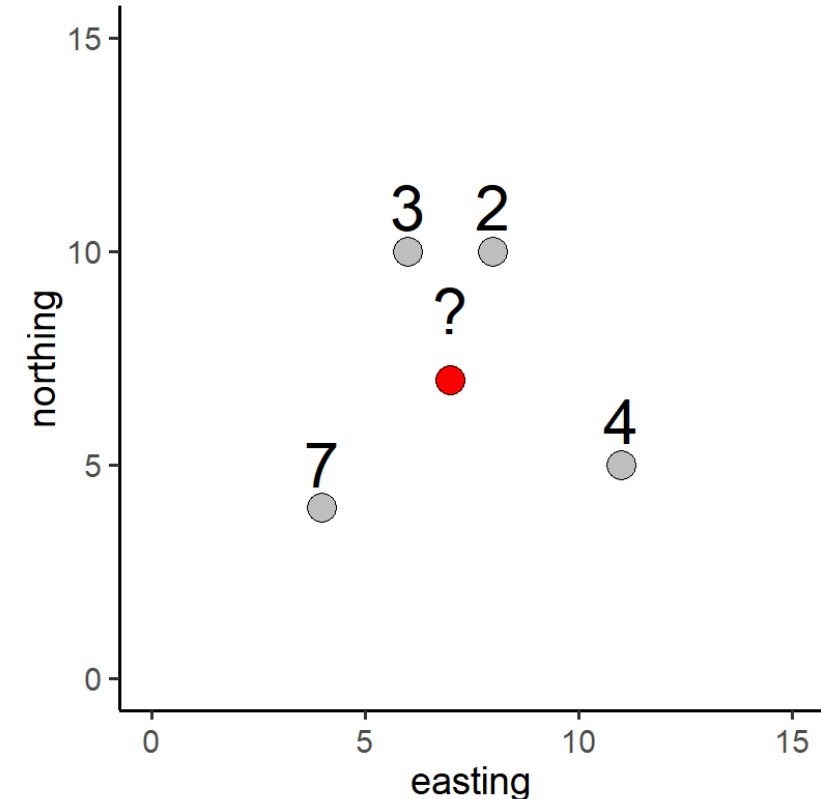
$$w_i = \frac{1}{d_{ij}^p}$$



Point ID	northing	easting	z	$Distance = d_{ij}$	$w_i = 1/d_{ij}$	$w_i * z$
1	11	5	4	4.47	0.22	0.89
2	6	10	3	3.16	0.32	0.95
3	8	10	2	3.16	0.32	0.63
4	4	4	7	4.24	0.24	1.65
Sum					1.09	4.13
5	7	7	?			

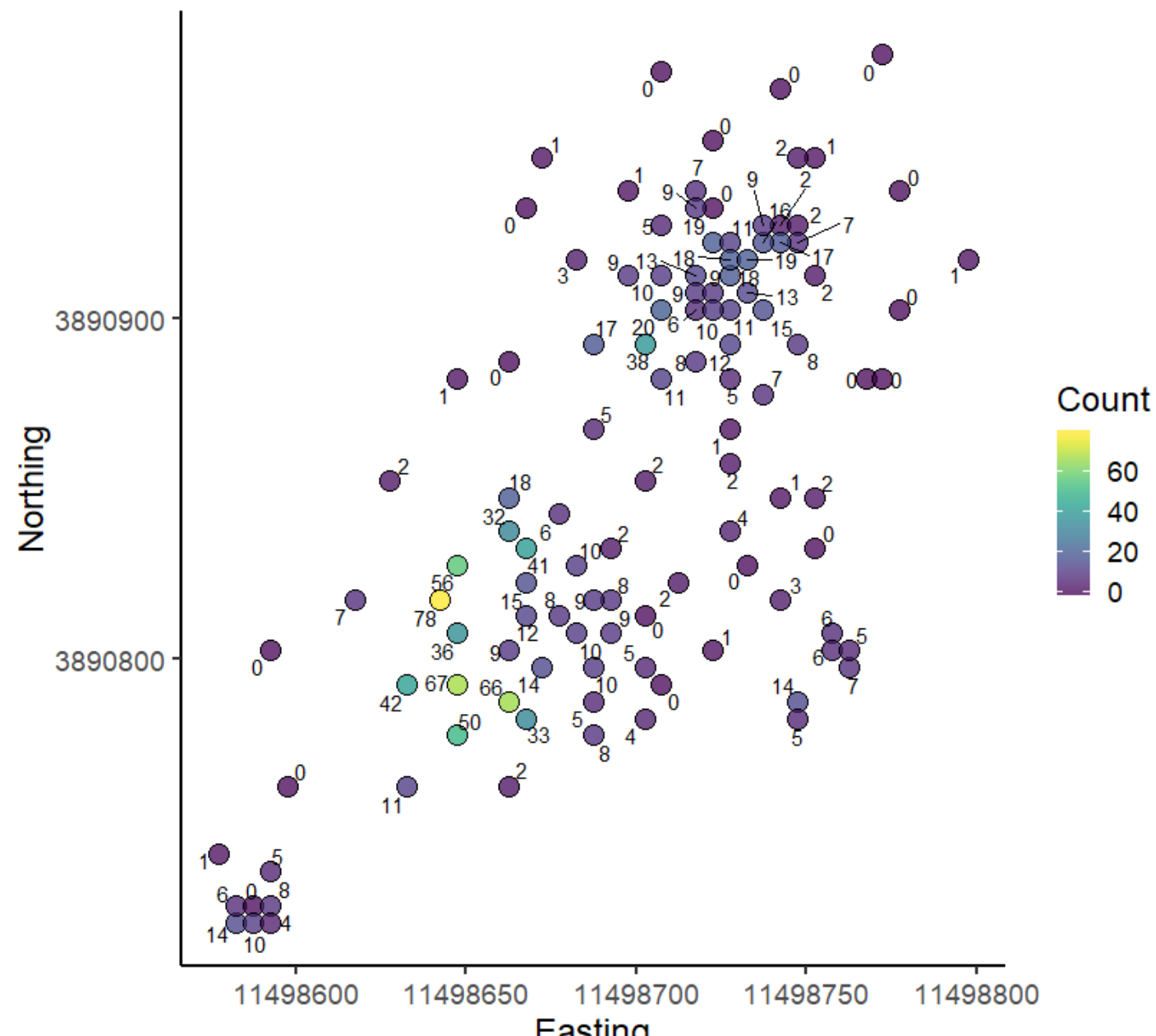
Inverse Distance Weighting

$$w_i = \frac{1}{d_{ij}^p}$$



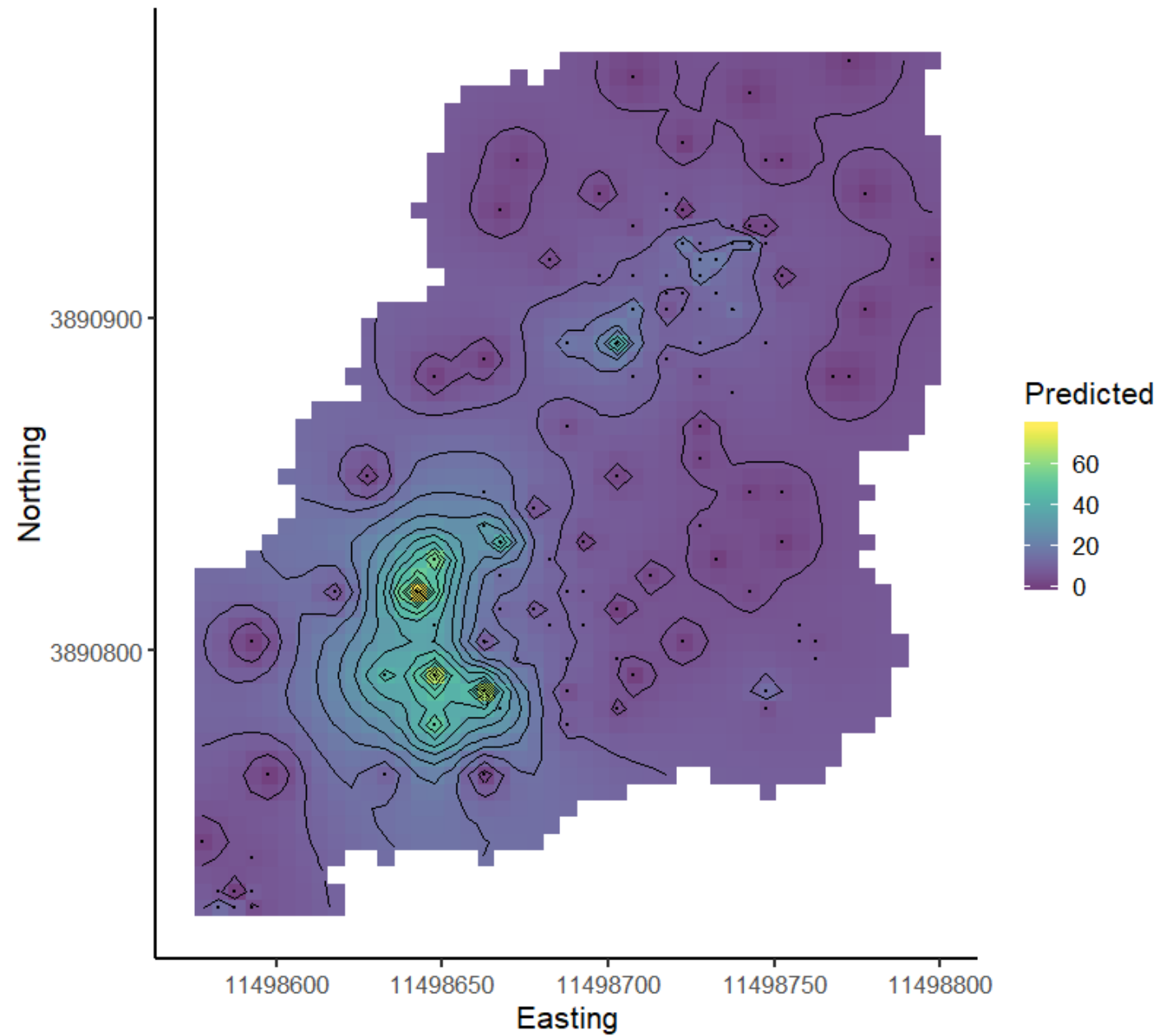
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<i>Sum</i>					1.09	4.13
5	7	7	<i>4.13/1.09 = 3.8</i>			

Creamware



Creamware

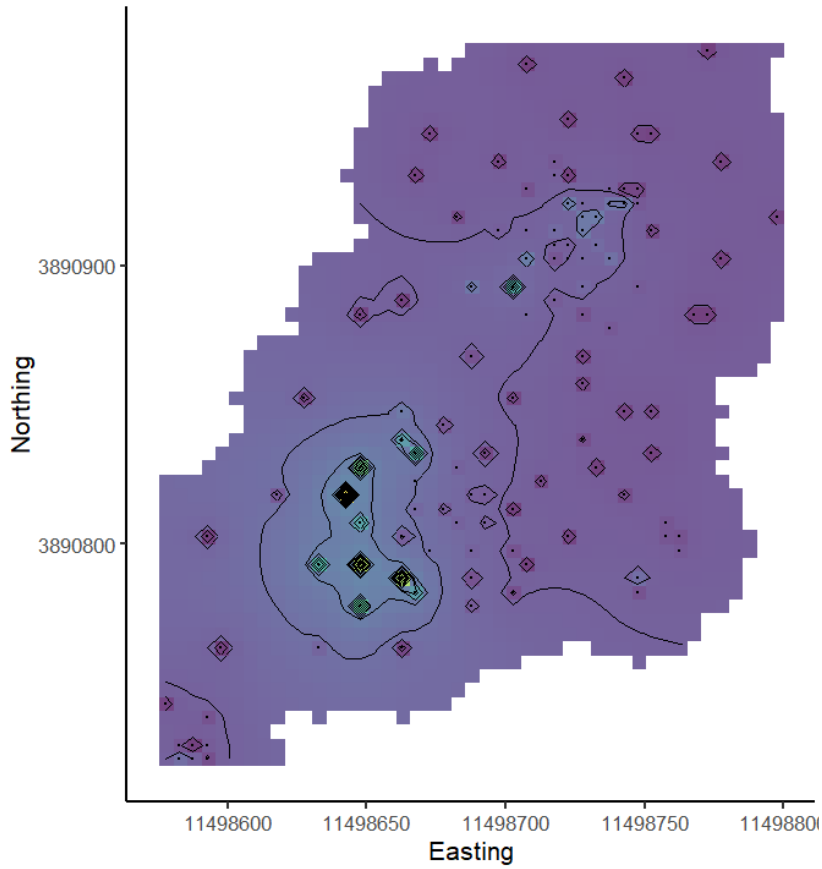
IDW, $p=2$



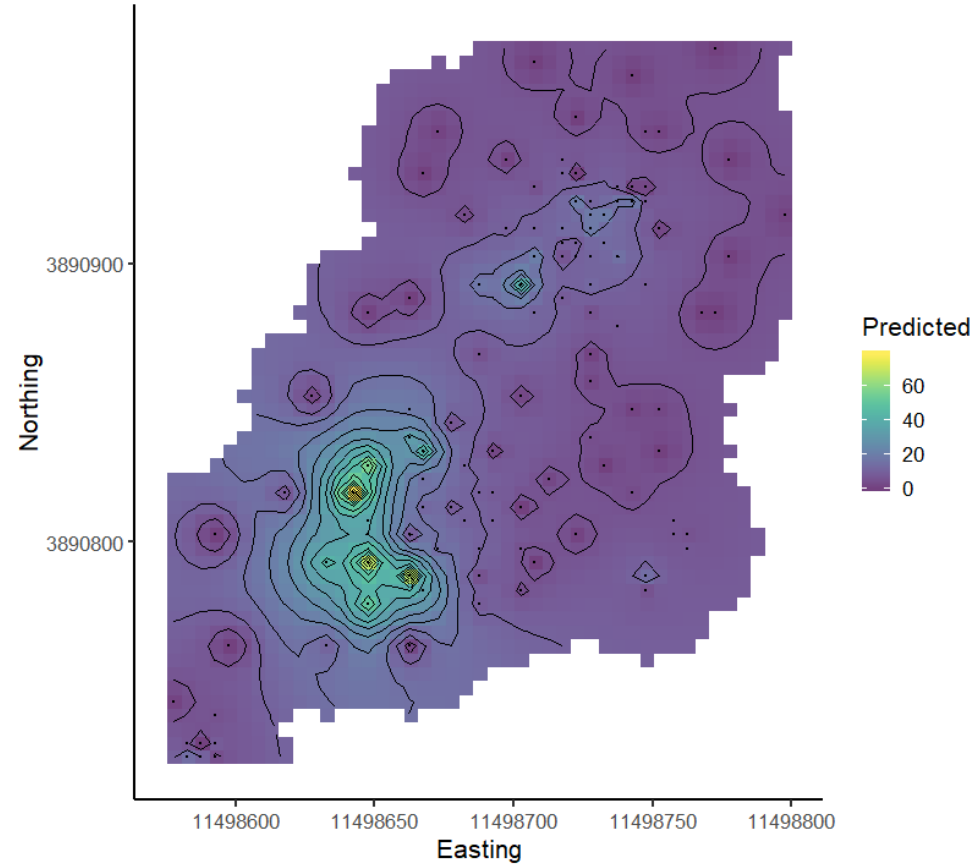
Pesky Questions about IDW

- what value for p ?

Creamware
IDW, $p=1$



Creamware
IDW, $p=2$



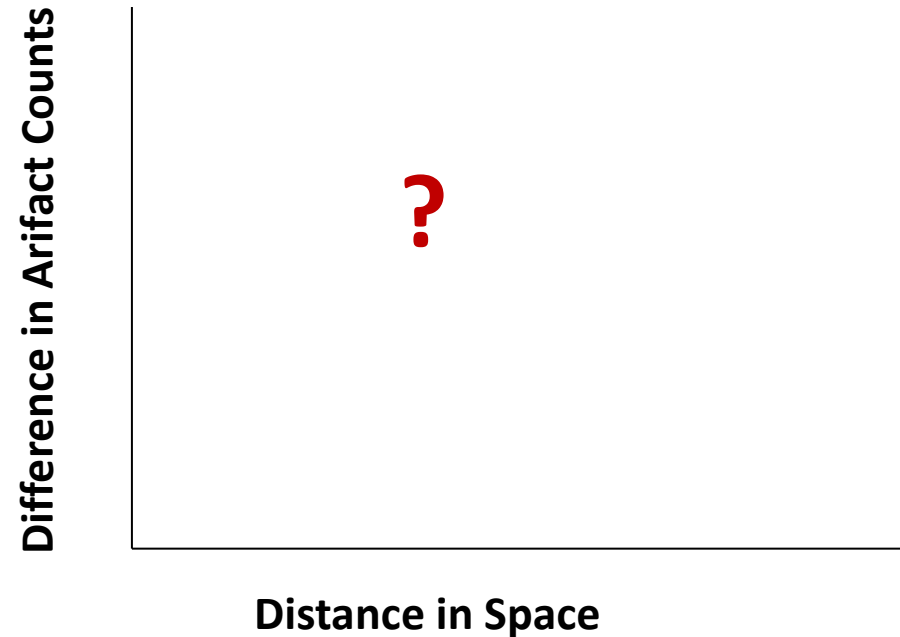
Higher values for p
tend to flatten out the
spikes near the known
points

Doing Better than IDW

*It would make sense if the weight we used for a z value at a given distance between a known point and the point we want to interpolate took into account the way in which expected **differences** between z-values increase with **distances** between x,y coordinates....*

“Spatial autocorrelation”

To what extent do quadrats that are farther apart in 2-d space (e.g. Easting and Northing) tend to have variable values (e.g. artifact counts) that are more different.



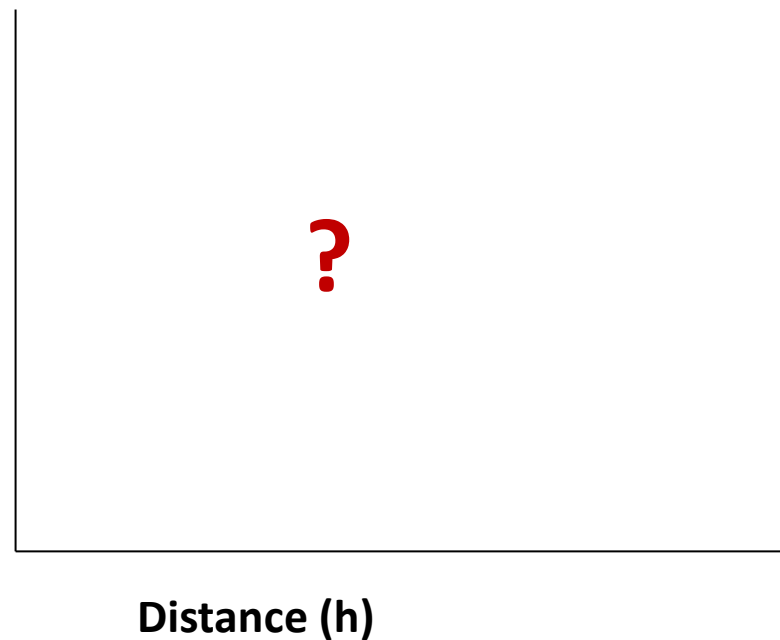
Kriging

A weighted-averaging interpolation method in which the ***weights depend on the spatial autocorrelation structure of the data***, AND that produces estimates of z values for unsampled x - y points that are designed to ***minimize mean-squared prediction error***.

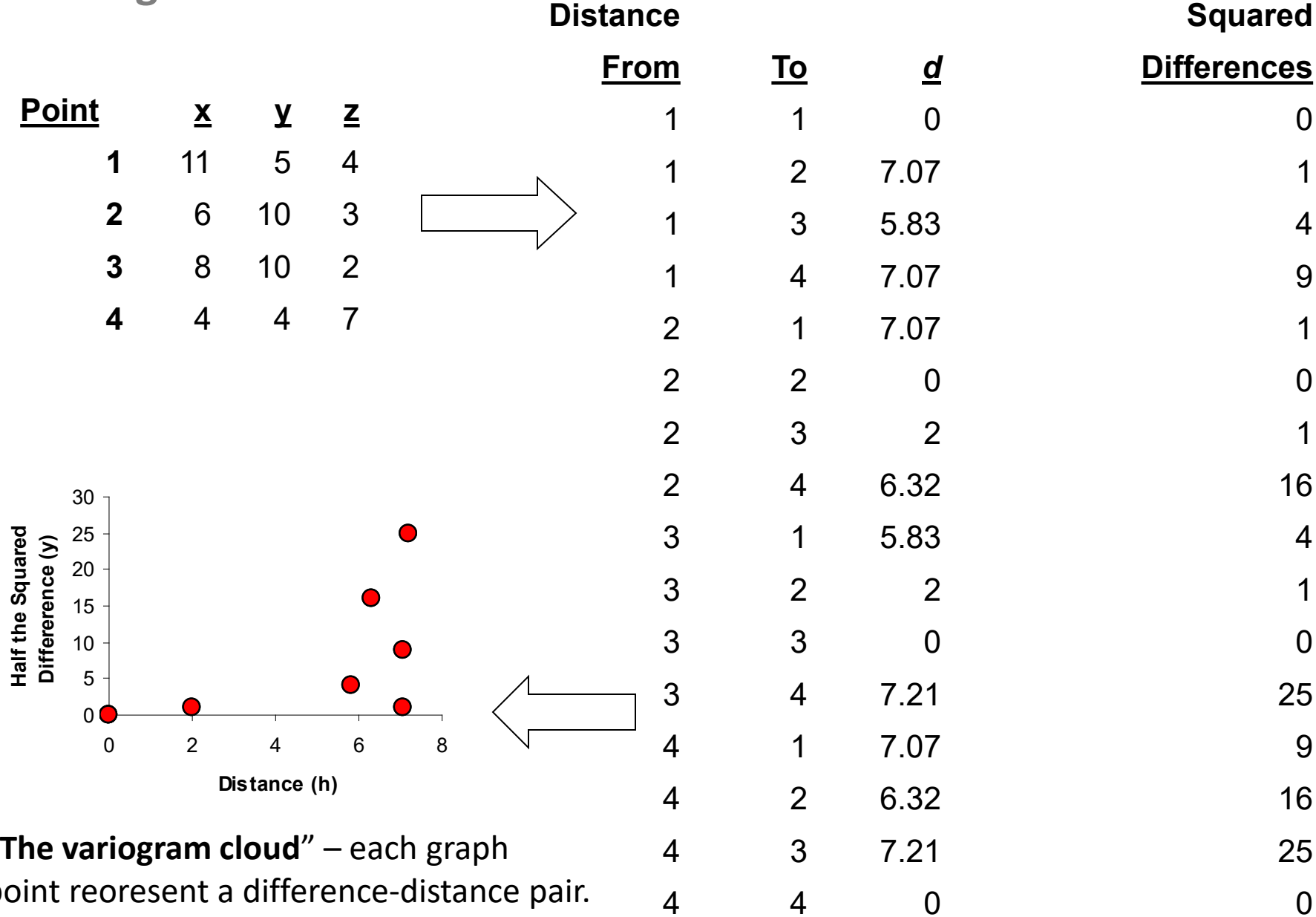
Variogram

The graphical tool we use to measure the autocorrelation structure of spatial data.

This is a Greek gamma: γ  Half the Squared Difference (γ)



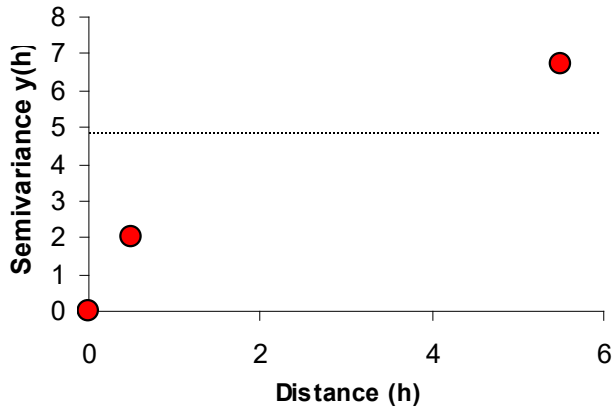
The Variogram



The Variogram

<u>Point</u>	<u>x</u>	<u>y</u>	<u>z</u>
1	11	5	4
2	6	10	3
3	8	10	2
4	4	4	7

Variance 4.67

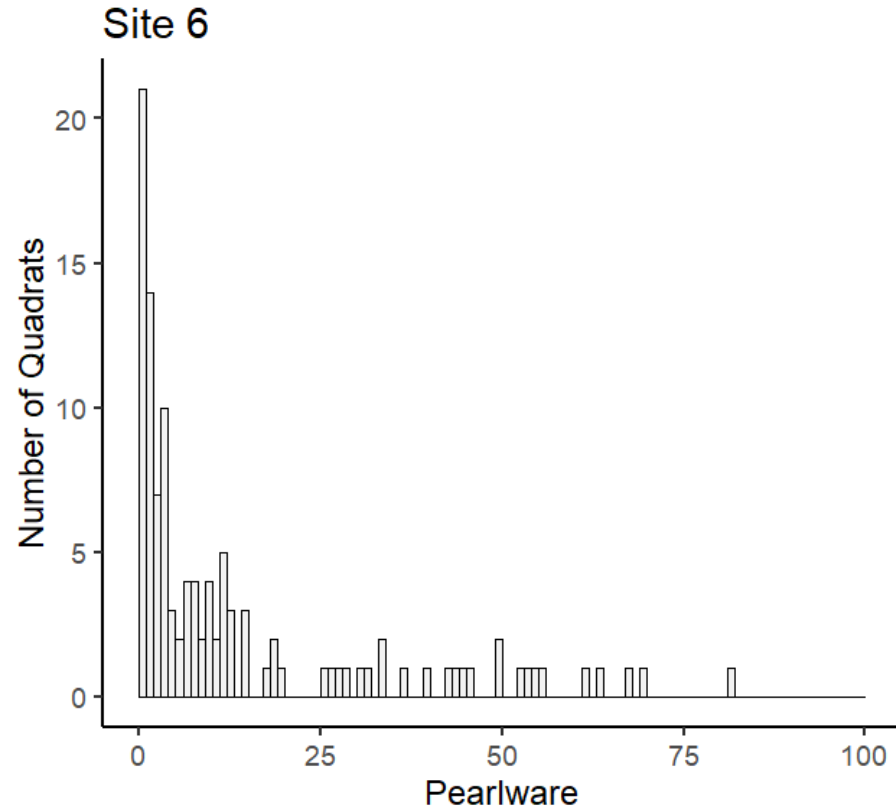
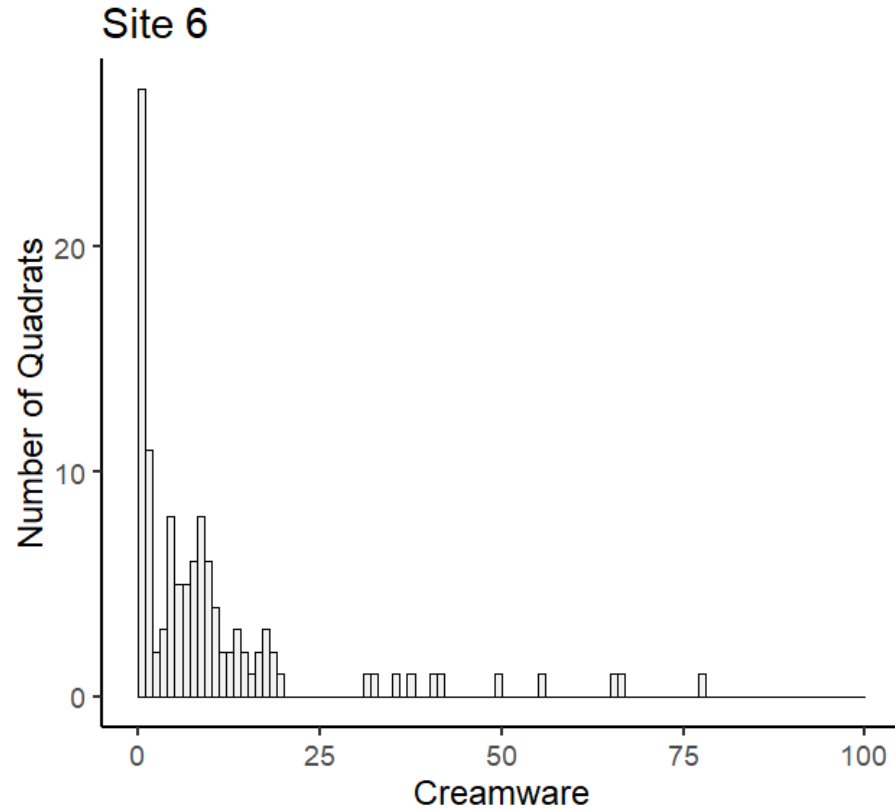


<u>Distance</u>	<u>From</u>	<u>To</u>	<u>h</u>	<u>Squared</u>	<u>Distance</u>	<u>y(h)</u>	<u>mean(h)</u>
				<u>Differences</u>	<u>Class</u>		
1	1	1	0	0	0		
2	2	2	0	0	0		
3	3	3	0	0	0		
4	4	4	0	0	0	0	0
2	3	2	2	1	1-5		
3	2	1	2	1	1-5	0.5	2
1	3	1	5.831	4	5-10		
3	1	2	5.831	4	5-10		
2	4	2	6.325	16	5-10		
4	2	1	6.325	16	5-10		
1	2	1	7.071	1	5-10		
1	4	3	7.071	9	5-10		
2	1	2	7.071	1	5-10		
4	1	3	7.071	9	5-10		
3	4	3	7.211	25	5-10		
4	3	2	7.211	25	5-10	5.5	6.70175

“The variogram” – each graph point represents the means of several difference-distance pairs.

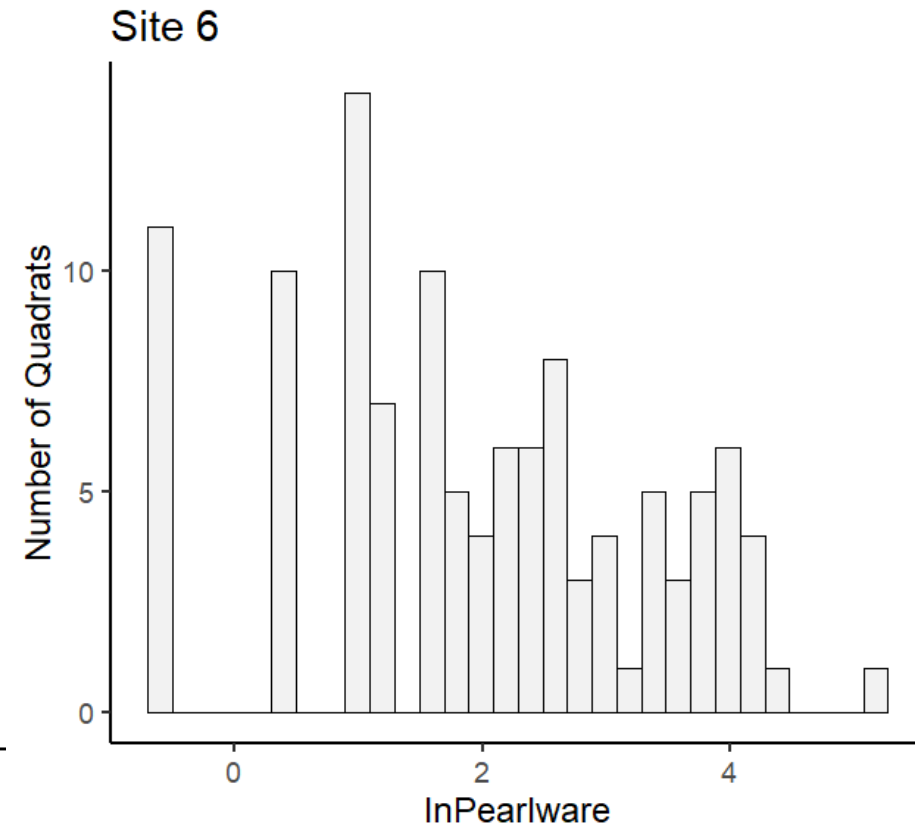
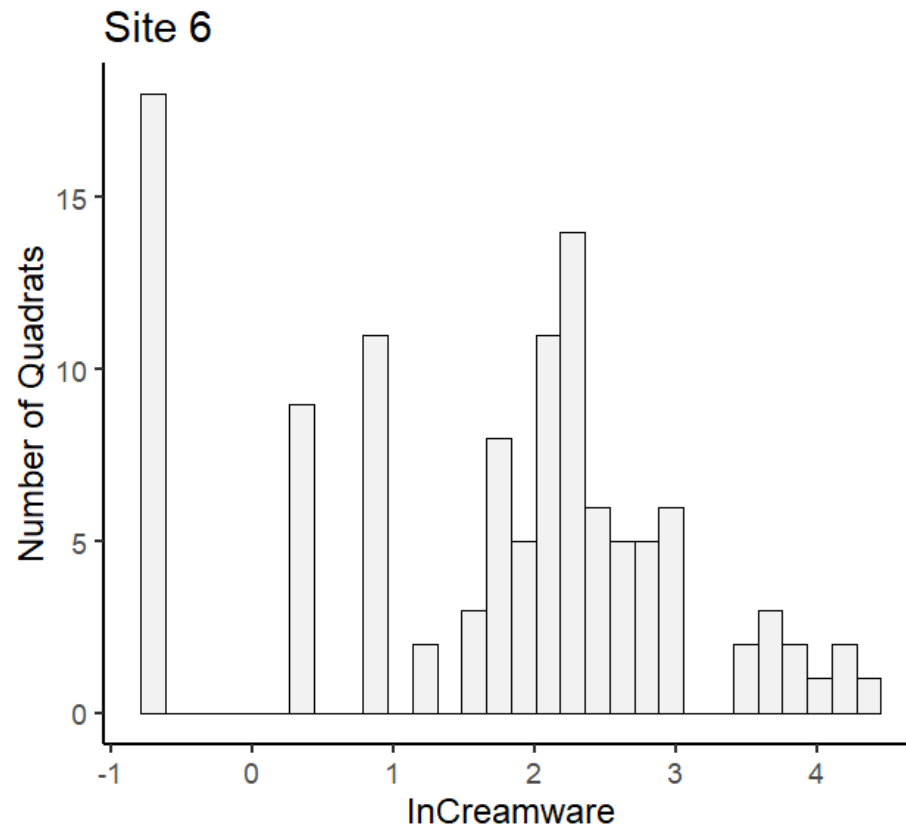
The Variogram

- The mathematical model behind the variogram and kriging assumes that the spatially distributed variable has a normal or Gaussian distribution.
- But artifact counts always have long right tails...



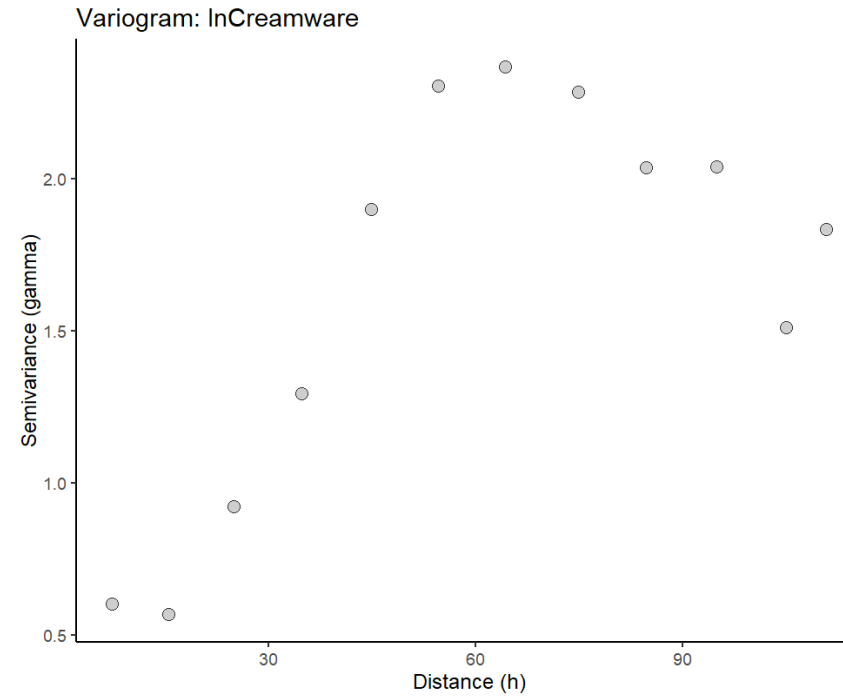
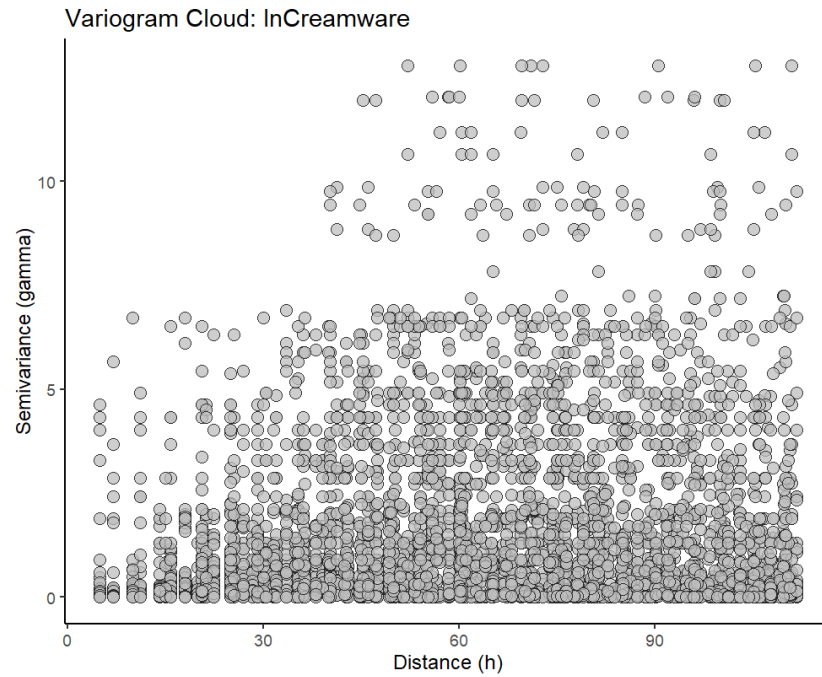
The Variogram

- Transforming the counts to a log scale helps.
- Because $\ln(0)$ is undefined, we take logs of “started counts”
 - e.g. $\ln(\text{Creamware} + .5)$



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- Because $\ln(0)$ is undefined, we take logs of “started counts”
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The Variogram

$$\gamma_h = \frac{1}{2n(h)} \sum_{i=1}^n (z(x_i, y_i) - z((x_i, y_i) + h))^2$$

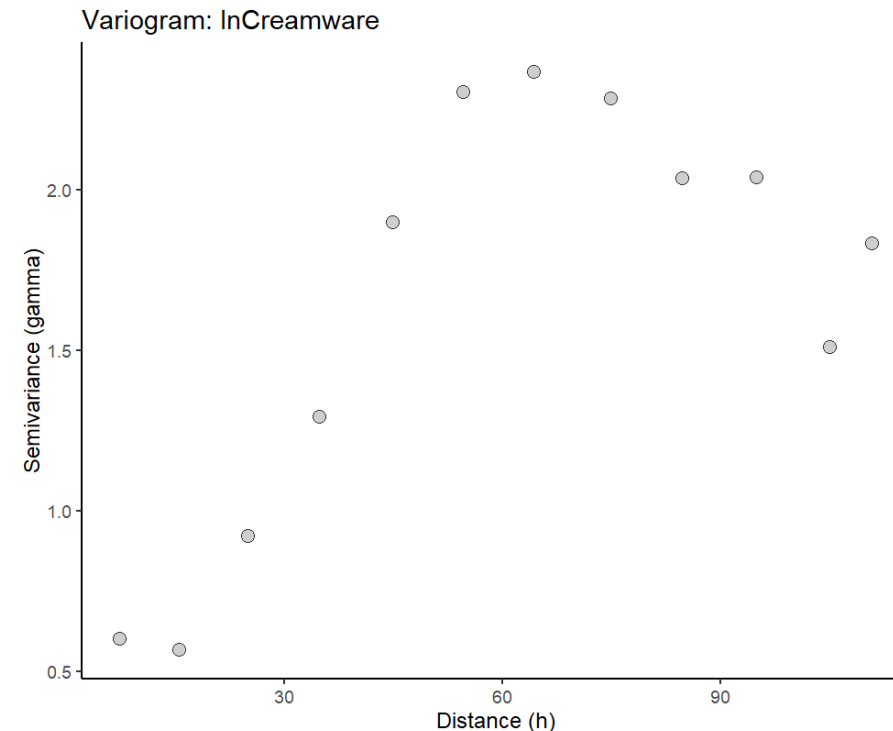
Annotations for the formula:

- γ_h : The semivariance for distance class h
- $2n(h)$: $n(h)$ = the number of point pairs that fall in distance class h
- $\sum_{i=1}^n$: Add up all the difference in pairs of z values
- $z(x_i, y_i)$: the value of z at the i 'th point with coordinates x, y
- $z((x_i, y_i) + h)$: $z(x_i, y_i) + h$ = the value of z at the i 'th point that is h away from the point with coordinates x, y
- $)^2$: Square the differences in pairs of z values

(x, y) are 2-d spatial coordinates (easting, northing)

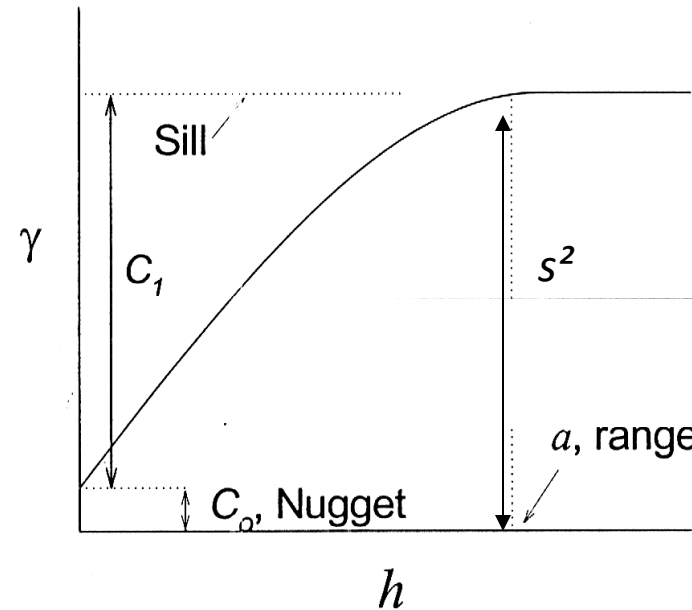
z is the variable value (artifact counts)

h is a distance and direction vector:
 “all the points that are a certain distance apart from the i 'th x, y pair”.



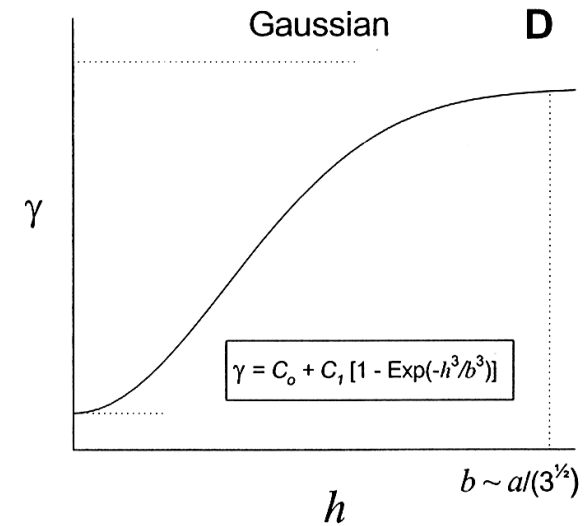
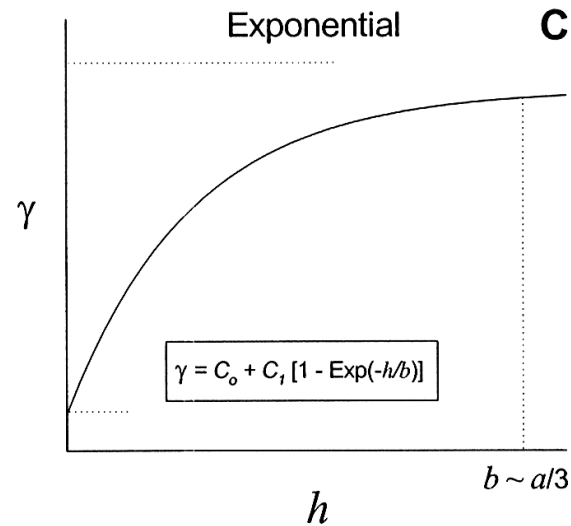
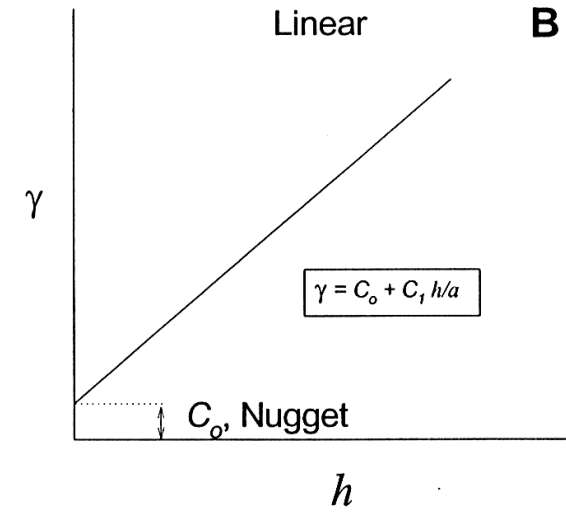
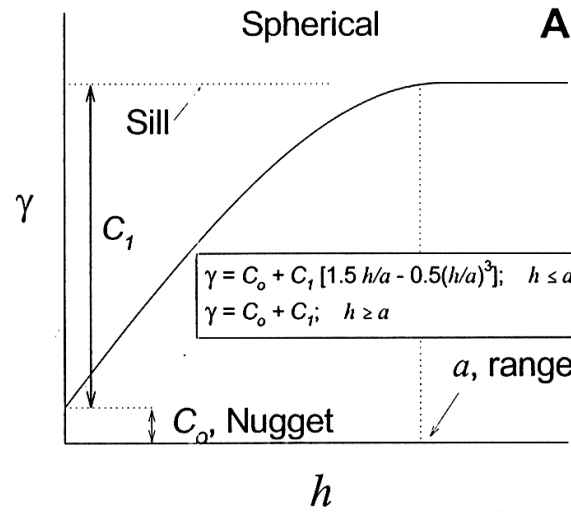
Variogram Lingo

- **Sill:** for larger values of h the variogram levels out, indicating that there no longer is any auto correlation between data points.
- *If the data are “well behaved” (Gaussian and stationary) the sill should be equal to the **variance** (s^2) of the z values.*
- **Range:** is the value of h where the sill occurs (or 95% of the value of the sill). This is the distance beyond which pairs of values are no longer autocorrelated.
- **Nugget variance:** a non-zero value for *gamma* when $h = 0$. Produced by various sources of unexplained error (*e.g.* measurement error).



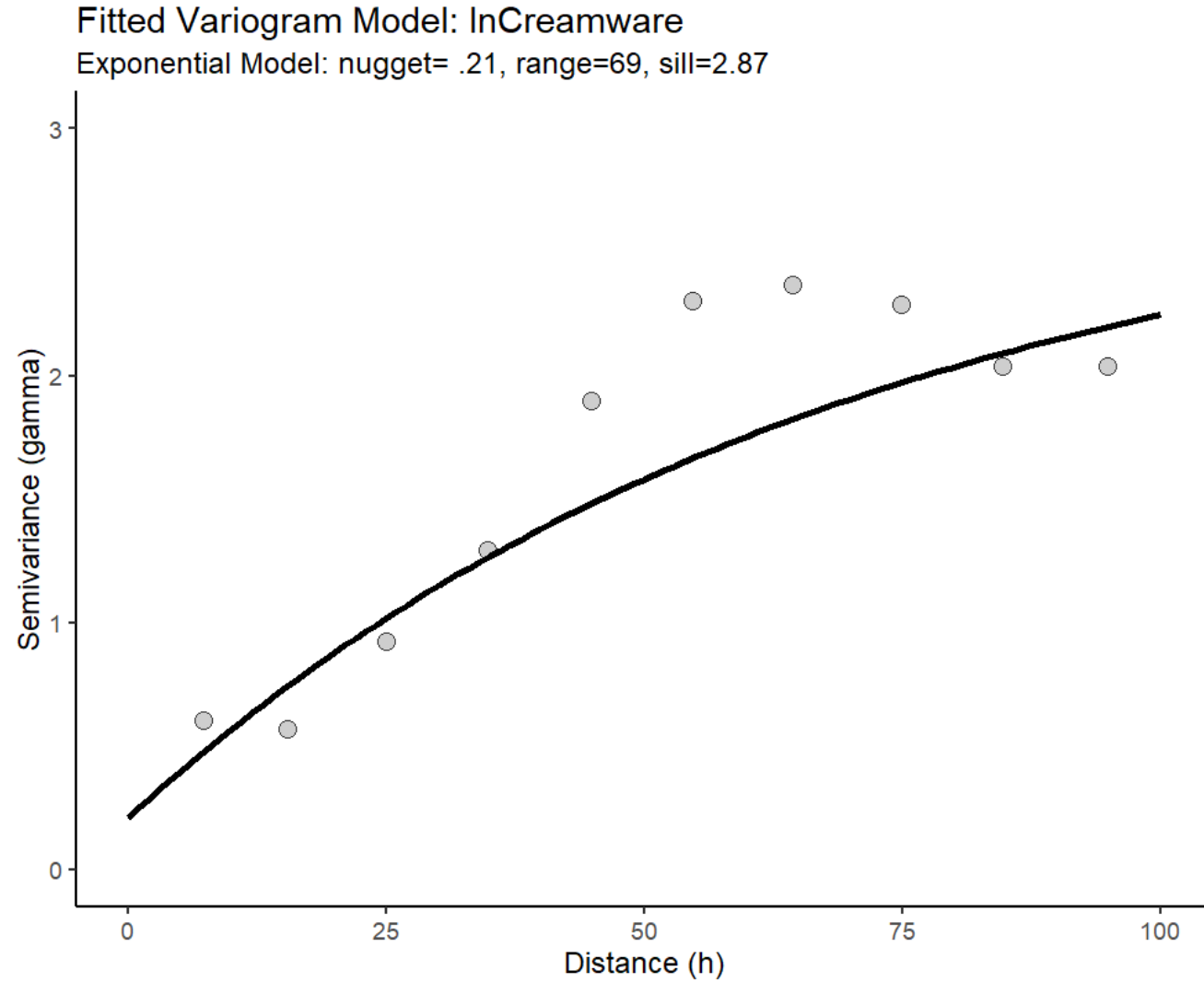
The Variogram

Variogram Models:
Differently shaped
curves, defined by
different equations.



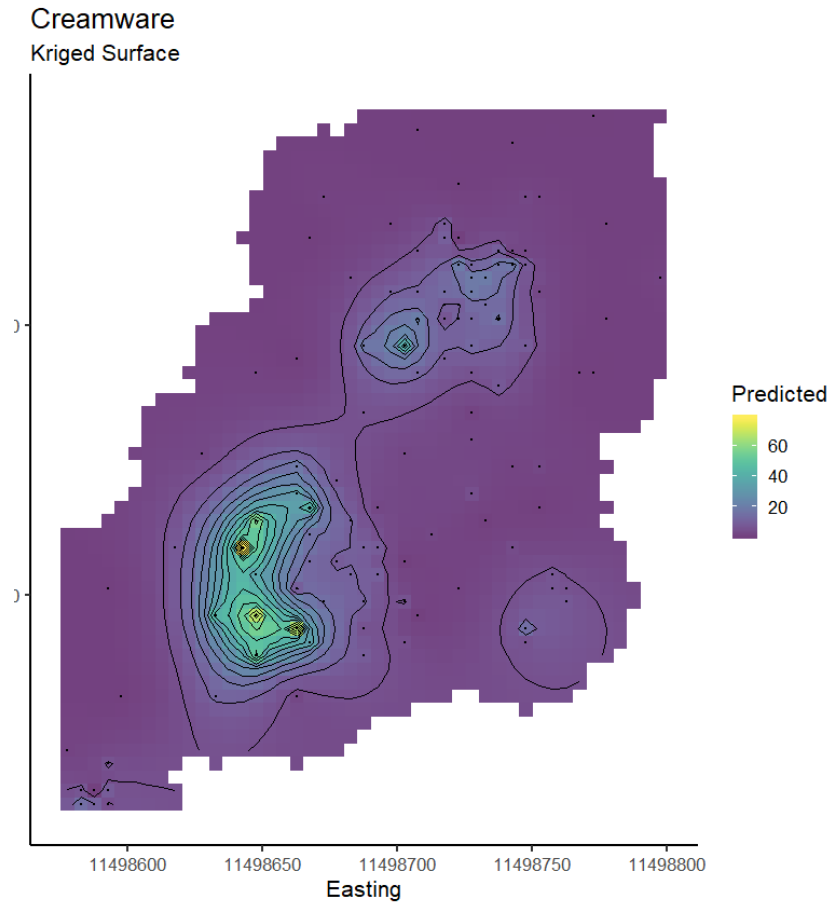
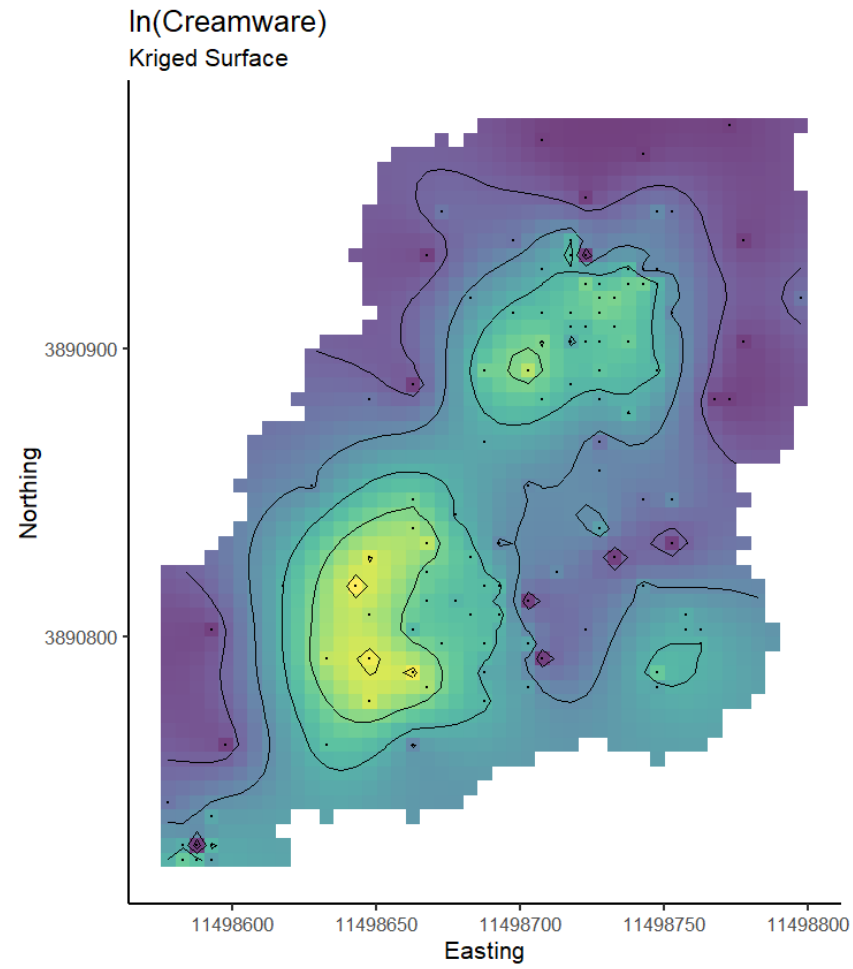
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Site 6

Kriged Density Surfaces

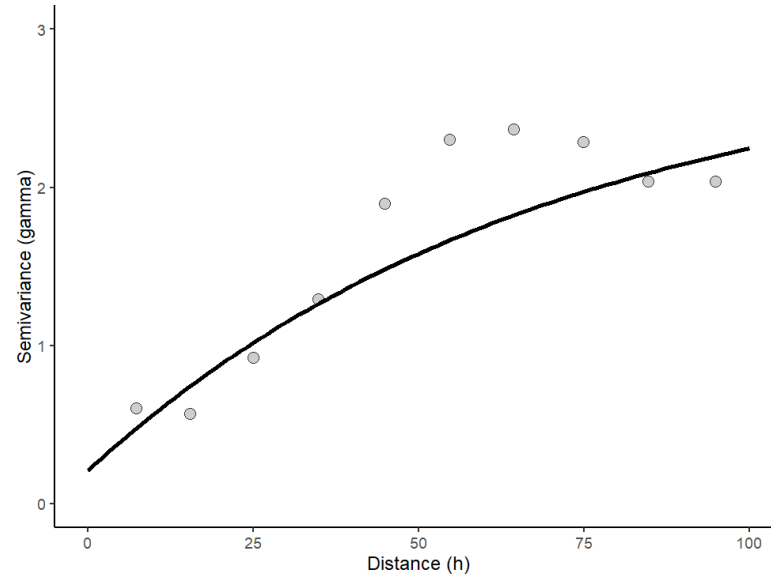


The variogram is a useful spatial data analysis tool !!

- You can use it during and excavation to see if your spatial sampling strategy is sufficient to capture accurately patterns in intra-site artifact distributions.
- Questions...
 - For which sites do we see positive spatial autocorrelation in the data?
 - For each site, what is the range (distance) over which counts are positively autocorrelated?

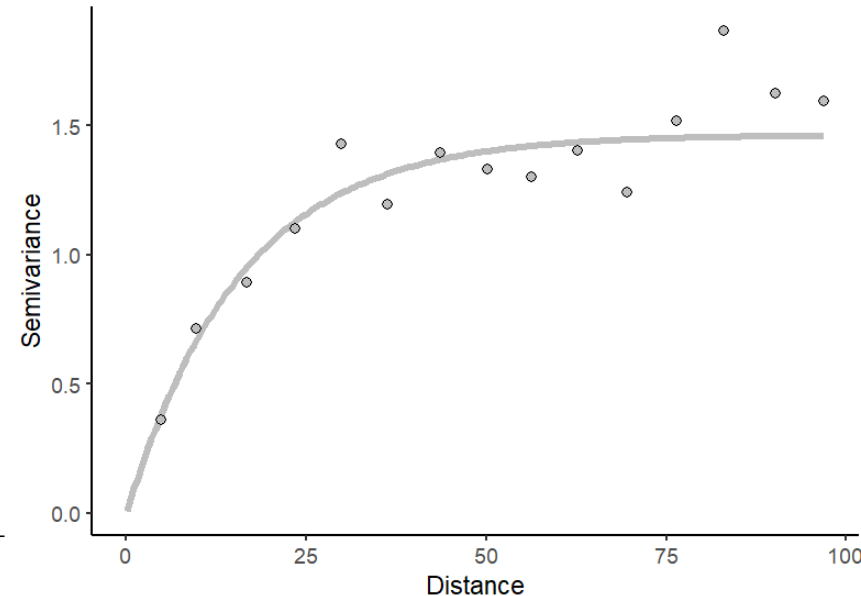
Site 6

Empirical Variogram with Fitted Model
Creamware: ln(started counts)



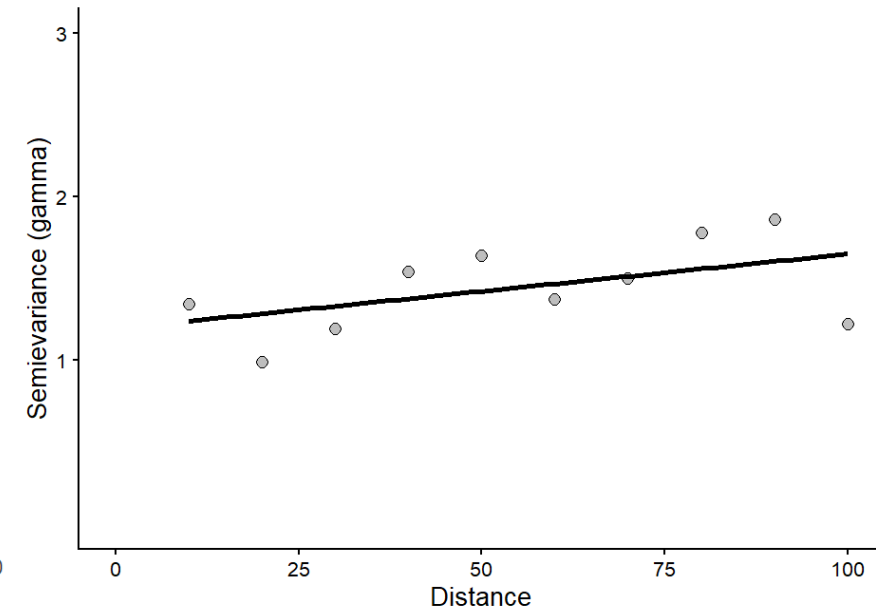
Site 30

Empirical Variogram with Fitted Model
Creamware: ln(started counts)



Site X

Empirical Variogram with Fitted Model

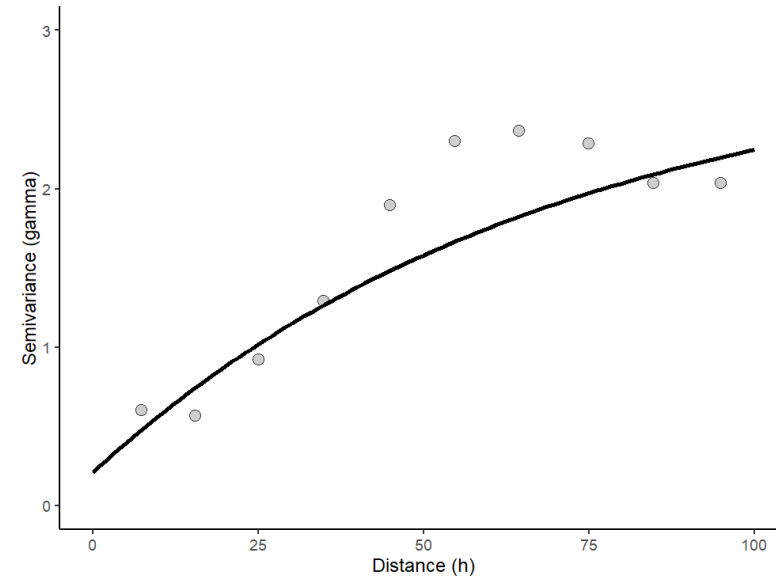


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Site 6

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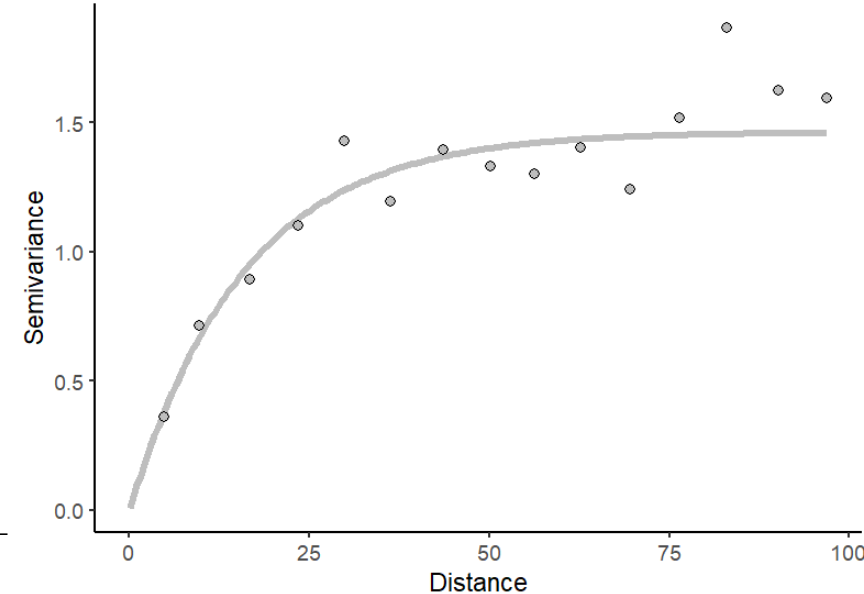


- Positive Autocorrelation
- Range ~ 50
- Interpolating z values at unsampled points justified.

Site 30

Empirical Variogram with Fitted Model

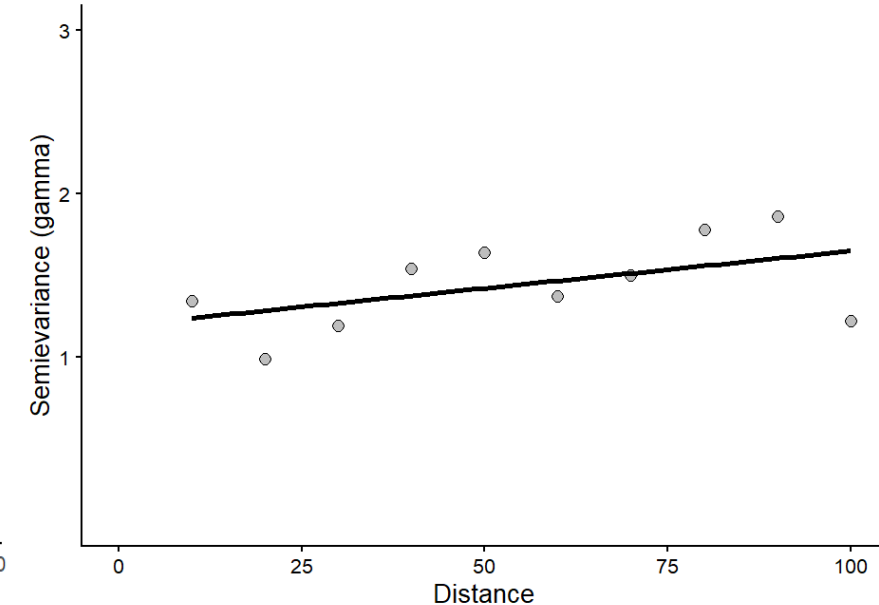
Creamware: ln(started counts)



- Positive Autocorrelation
- Range ~ 30
- Interpolating z values at unsampled points justified.

Site X

Empirical Variogram with Fitted Model



- No Autocorrelation. The surface is “white noise”.
- Range < 10
- Interpolating z values at unsampled points is NOT justified. Given their size, the quadrats are too far apart. You need closer quad spacing.